Université de Genève

Départment d'informatique

Éléments de la Théorie de l'information

Information Theory for Data Science and Machine Learning — 10.03.2021

Problem Set 2 — Due: 16.03.2021

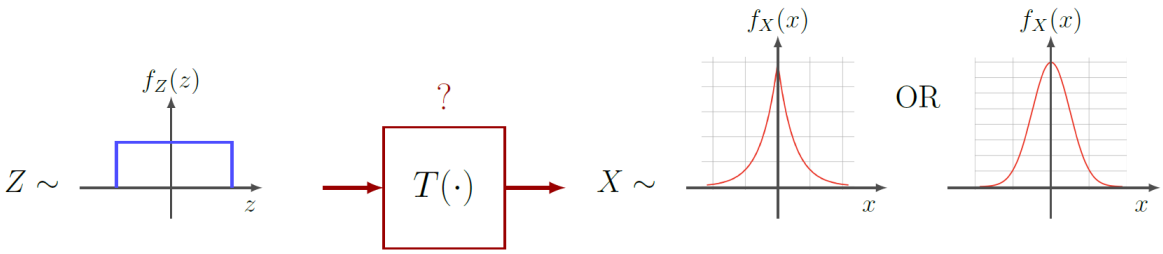
Note: Please follow the instruction on `https://moodle.unige.ch/mod/page/view.php?id=171850`.

Problem 1 (Transformation of Random Variables—40 points).

a) Given a 1-dimensional uniform Random Variable (RV) $Z \sim f_Z(z) = \mathcal{U}(-a, a)$, derive the analytical transformation $X = T(Z)$ for a Gaussian RV $X \sim f_X(x) = \mathcal{N}(0, \sigma^2)$ and a Laplacian RV $X \sim f_X(x) = \mathcal{L}(0, b)$, where:

$$\begin{aligned}
\mathcal{U}(x|a,b) &= \begin{cases} \frac{1}{b-a}, & \forall x \in [a,b] \\ 0, & \text{otherwise} \end{cases}, \\
\mathcal{N}(x|\mu,\sigma^2) &= \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \forall x \in \mathbb{R}, \\
\mathcal{L}(x|\mu,b) &= \frac{1}{2b} \exp\left(-\frac{|x-\mu|}{b}\right), \forall x \in \mathbb{R}.
\end{aligned}$$

The figure below explains the expected transformation.



Provide $T(\cdot)$ for each case.

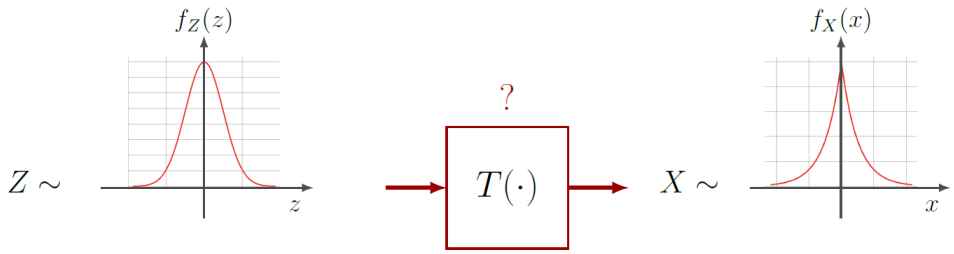
Hint: You have seen that the conservation of probabilities $\int f_Z(z)\mathrm{d}z = \int f_X(x)\mathrm{d}x$ implies

$$\int_a^b f_Z(z)\mathrm{d}z = \int_{T(a)}^{T(b)} f_X(x)\mathrm{d}x. \tag{1}$$

In particular, this holds for

$$\int_{-\infty}^b f_Z(z)\mathrm{d}z = \int_{-\infty}^{T(b)} f_X(x)\mathrm{d}x. \tag{2}$$

Note that $T(-\infty)$ isn't necessarily equal to $-\infty$ and may even not be defined, but in any case $f_Z(z) = f_X(x) = 0$ from $-\infty$ to the lower element of their respective domain. Both expressions in Eq. (2) are the definitions of the Cumulative Distribution Functions (CDF) $F_Z(b) = \Pr[Z \leq b]$ and $F_X(T(b)) = \Pr[X \leq T(b)]$ respectively.



b) Derive the analytical transform of a Gaussian into a Laplacian PDF.

What is the transformation $X = T(Z)$?

Problem 2 (**Effect of Randomization—20 points**).

In this problem, our goal is to investigate the effect of adding independent noise $\mathbf{Z}$ to a correlated signal $\mathbf{X}$. To this end, generate a random vector

$$\mathbf{X} \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & 0.8 \\ 0.8 & 1 \end{bmatrix}\right)$$

Then add the independent random noise

$$\mathbf{Z} \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} \sigma_{\mathbf{z}}^2 & 0 \\ 0 & \sigma_{\mathbf{z}}^2 \end{bmatrix}\right)$$

to $\mathbf{X}$. Now, investigate the effect of $\sigma_{\mathbf{z}}^2$ on the 2-dimensional pdf of the random vector $\mathbf{Y} = \mathbf{X}+\mathbf{Z}$. Investigate your results for $\sigma_{\mathbf{z}}^2 \in \{0.1, 0.5, 1, 2, 10\}$ by plotting $1'000$ realizations, i.e. randomly generated vectors, for $\mathbf{X}$ and $\mathbf{Y}$. Describe your observations and conclusions. Indicate $\sigma_{\mathbf{z}}^2$ after which we do not observe correlation properties in $\mathbf{Y}$ any more.

Problem 3 (**Markov's Inequality and Chebyshev's Inequality—40 points**).

a) (Markov's inequality) For any non-negative random variable X and any $t > 0$, show that

$$\Pr\{X \geq t\} \leq \frac{\mathbb{E}[X]}{t}.$$

Give an example of a random variable that saturates this inequality, i.e. that satisfies the strict equality.

b) (Chebyshev's inequality) Let Y be a random variable with mean μ and variance σ^2. By letting $X = (Y - \mu)^2$, show that for any $\epsilon > 0$,

$$\Pr\{|Y - \mu| \geq \epsilon\} \leq \frac{\sigma^2}{\epsilon^2}.$$

c) (Weak law of large numbers) Let $Z_1, Z_2, ..., Z_n$ be a sequence of i.i.d. random variables with mean μ and variance σ^2. Let $\bar{Z}_n = \frac{1}{n}\sum_{i=1}^{n} Z_i$ be the sample mean. Show that:

$$\Pr\{|\bar{Z}_n - \mu| \geq \epsilon\} \leq \frac{\sigma^2}{n\epsilon^2}.$$

Thus, $\Pr\{|\bar{Z}_n - \mu| \geq \epsilon\} \to 0$ as $n \to \infty$. This is known as the weak law of large numbers.